

# Axworthy Ghost Pulley System — User Guide

Firmware: Block 2 (XIAO SAMD21 port), based on v1.3

This guide covers everything from first-time setup to day-to-day operation and fine-tuning. It is written for the person operating the ghost, not just the programmer. No coding knowledge is required to use the device once it is flashed.

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## 1. What this device does

The Ghost Pulley System drives a stepper motor that pulls a prop ("the ghost") back and forth along a line (a pulley/cable run). You teach it two end positions, and it travels between them automatically, with smooth acceleration and deceleration, pausing (dwelling) at each end. It can run continuously or on a daily schedule (for example, only between 6 PM and 11 PM).

Everything is controlled from a small OLED screen and a **single rotary knob that also clicks** (a rotary encoder with a push-button). There is also a **limit switch** at the home end for safety and homing.

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## 2. Hardware overview

| Part                                   | Role                                       |
|----------------------------------------|--------------------------------------------|
| Seeeduino XIAO SAMD21                  | The microcontroller ("the brain")          |
| A4988 driver                           | Powers the stepper motor                   |
| NEMA 17 stepper motor                  | Moves the ghost                            |
| SSD1309 / SSD1306 2.42" OLED (I2C)     | The 128×64 display                         |
| EC11 rotary encoder (with push button) | Your only control: turn + click            |
| DS3231 RTC                             | Real-time clock, for scheduling            |
| Limit switch (home end)                | Safety stop + finds "home" (position zero) |

### Pin map (XIAO SAMD21)

|                   |                    |                      |
|-------------------|--------------------|----------------------|
| D0 = STEP         | D1 = DIR           | D2 = ENABLE          |
| D3 = ENC_CLK      | D6 = ENC_DT        | D7 = ENC_SW (button) |
| D4 = SDA          | D5 = SCL           | D8 = LIMIT switch    |
| D9 = AUX1 (spare) | D10 = AUX2 (spare) |                      |

## Important hardware notes

- **Microstepping is set by an on-board DIP switch, not the firmware.** The firmware's distance/speed math assumes **1/8 microstepping (MS1/MS2/MS3 = High/High/Low)**. If the DIP switch is set differently, all distance and speed readings will be wrong. Leave it at 1/8 unless you also change the firmware constant STEPS\_PER\_INCH (339.4 for 1/8; 169.7 for 1/4; 678.8 for 1/16).
  - The mechanical assumption is a **76.2 mm diameter pulley** with a **2:1 gear**, giving 3200 steps per output revolution and **339.4 steps per inch** of travel.
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## 3. First-time setup (software)

You only need to do this once, to get the firmware onto the board.

1. **Install the Arduino IDE** (if you don't already have it).
2. **Add the board support:** In the Arduino IDE, open Tools > Board > Boards Manager, search for "**Seeed SAMD Boards**", and install it. Then select Tools > Board > Seeeduino XIAO.
3. **Install the required libraries** via Tools > Manage Libraries:
  - Adafruit SSD1306
  - Adafruit GFX Library
  - FlashStorage by cmaglie
  - RTCLib by Adafruit
4. **Open** ghost\_pulley\_xiao.ino.
5. **Plug in** the XIAO via USB, pick its port under Tools > Port, and click **Upload**.

### Optional: run without the OLED (debugging)

Near the top of the file there is a line:

```
//#define NO_OLED // uncomment to use Serial Monitor instead of OLED
```

Remove the // to build a version that prints all menus and status to the **Serial Monitor** (9600 baud) instead of the OLED. Useful if the display isn't wired yet. Leave it commented for normal use.

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## 4. The control: one knob, two actions

Everything is done with the rotary encoder:

- **Turn** = move the selection highlight, or change a value.
- **Click (short press)** = select / enter / confirm.

That's it. There is no separate keypad. Clockwise generally means "down / increase," counter-clockwise means "up / decrease."

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## 5. First power-on: what happens automatically

When the device boots:

1. It shows a brief splash screen.
2. It **homes** the motor — it slowly drives toward the limit switch until the switch triggers, backs off slightly, and calls that spot **position zero (home)**. The ghost cannot travel "behind" home.
3. If **no positions have been taught yet**, it jumps straight into **Learn mode** (see Section 7) so you can set them. If positions were previously saved, it goes to the main menu instead.

All your settings and taught positions are stored in flash and survive power loss, so this full "teach" only needs to happen once (or whenever you want to change the travel range).

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## 6. The main menu

The main menu has five lines. Turn to highlight (>), click to activate.

GHOST PULLEY

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> Run: Off  
Mode: Fwd-Back  
Motion  
Setup  
Diagnostics

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Off 7:30PM <- status bar (state + clock)

| Item               | What clicking it does                                                                   |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Run</b>         | Cycles the run state: <b>Off</b> → <b>Schedule</b> → <b>Continuous</b> → <b>Off ...</b> |
| <b>Mode</b>        | Toggles the motion style: <b>Fwd-Back</b> ↔ <b>Wander</b>                               |
| <b>Motion</b>      | Opens the Motion settings sub-menu (speed, accel, dwell)                                |
| <b>Setup</b>       | Opens the Setup sub-menu (Learn, Schedule, Clock)                                       |
| <b>Diagnostics</b> | Opens a live position/velocity readout                                                  |

The **status bar** at the bottom always shows the current state and time:

- Off 7:30PM — not running.
- Running 7:30PM — actively traveling.

- Starting 7:30PM — Continuous mode, about to move.
  - Next: 6:00PM 7:30PM — Schedule mode, waiting for the next start time.
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## 7. Teaching the two end positions (Learn mode)

This is the most important setup step. It defines where the ghost travels **between**. Reach it via Setup > Learn Positions (or it starts automatically on first boot).

Learn mode runs in two stages: **Position A**, then **Position B**.

### How to jog and set a position:

1. The screen shows LEARN POS A, the current position in feet, and the direction indicator (STOPPED / FORWARD / REVERSE).
2. **Turn clockwise** to jog the ghost **forward** (away from home).
3. **Turn counter-clockwise** to jog **reverse** (back toward home).
  - Reverse is blocked at home (position zero) — it cannot go negative.
4. Jog until the ghost is exactly where you want end **A**.
5. **Click** to save Position A. It immediately switches to LEARN POS B.
6. Repeat: jog to where you want end **B**, then **click** to save.
7. It confirms "Positions saved!" and returns to the main menu.

Both positions are stored permanently. You can re-teach them any time from Setup > Learn Positions.

Tip: A and B can be set in either order along the line; the system figures out which is nearer home. The travel range is simply the span between them.

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## 8. Running the ghost

Highlight **Run** on the main menu and click to cycle through:

- **Off** — motor disabled, ghost parked. (If it was running, it stops.)
- **Schedule** — runs only during the daily window you set (see Section 10). Outside the window it waits; the status bar shows the next start time.
- **Continuous** — runs all the time, back and forth, until you set it to Off.

If you try to select Schedule or Continuous **before** teaching positions, it refuses and shows "Set positions first!"

### What a run looks like

1. The motor enables and moves to Position A.
2. It **dwells** (pauses) for the dwell time.

3. It travels to the other position, dwells again, and repeats.
4. Each leg has smooth acceleration at the start and deceleration at the end.

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## 9. Motion modes

Set via the **Mode** item on the main menu.

- **Fwd-Back** — Predictable ping-pong:  $A \rightarrow B \rightarrow A \rightarrow B$ , pausing a **fixed** dwell time at each end.
- **Wander** — Spooky/random: it picks random target points between A and B, with a **random** dwell time between a min and max you set. Each move is forced to be at least long enough to complete its accel + decel ramp (no tiny twitches).

Switching mode **while the ghost is mid-move** is deferred — it finishes the current leg first, then switches (you'll see "Mode switch after move"). The two modes expose slightly different settings in the Motion menu (below).

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## 10. Settings reference and how to tweak them

Motion menu (main menu > Motion)

Turn to highlight a setting, **click to start editing** (the marker changes to \*), **turn to change the value**, then **click again to save**. Click << Back to return.

The settings list changes depending on the current Mode:

**Fwd-Back mode shows:**

| Setting     | Range    | Units / meaning                           |
|-------------|----------|-------------------------------------------|
| Speed in/s  | 0.5–45.0 | Cruise speed in inches/sec (shown as X.X) |
| Accel (sec) | 0.1–5.0  | Time to ramp up to speed                  |
| Decel (sec) | 0.1–5.0  | Time to ramp down to a stop               |
| Dwell (sec) | 0–30     | Fixed pause at each end                   |

**Wander mode shows:**

| Setting         | Range    | Units / meaning            |
|-----------------|----------|----------------------------|
| Speed in/s      | 0.5–45.0 | Cruise speed in inches/sec |
| Accel (sec)     | 0.1–5.0  | Ramp-up time               |
| Decel (sec)     | 0.1–5.0  | Ramp-down time             |
| Min Dwell (sec) | 0–30     | Shortest random pause      |
| Max Dwell (sec) | 1–60     | Longest random pause       |

Notes:

- **Speed** is displayed in inches/second but stored internally as motor steps/second. The conversion assumes 1/8 microstepping (see Section 2).
- **Accel/Decel** are *times*, not distances. If a move is too short to reach full speed within the ramp times, the firmware automatically shrinks both ramps proportionally so the move still completes smoothly.
- Every saved change is written to flash immediately.

## Setup menu (main menu > Setup)

| Item                   | What it does                               |
|------------------------|--------------------------------------------|
| << <b>Back</b>         | Return to main menu                        |
| <b>Learn Positions</b> | Re-teach end positions A and B (Section 7) |
| <b>Set Schedule</b>    | Set the daily start and stop time          |
| <b>Set Clock</b>       | Set the real-time clock                    |

### Set Schedule

Opens a four-step picker. At each step you **turn to change the value** and **click to advance** to the next:

1. Start hour
2. Start minute (5-minute increments)
3. Stop hour
4. Stop minute (5-minute increments)

After the fourth click it saves and returns. Times display in 12-hour AM/PM. A schedule that **wraps past midnight** is supported (for example, start 10:30 PM, stop 6:00 AM). Scheduling requires the RTC to be present; if it isn't, you'll see "RTC not found!"

### Set Clock

A three-step picker for the current time:

1. Turn to set the **hour**, click.
2. Turn to set the **minute**, click.
3. Turn to toggle **AM/PM**, click to save.

Set this correctly or the schedule will trigger at the wrong times.

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## 11. Diagnostics screen

main menu > Diagnostics shows a live readout (refreshes a few times per second). Click the button to return.

| Field         | Meaning                                                                   |
|---------------|---------------------------------------------------------------------------|
| <b>Pos</b>    | Current position from home, in feet                                       |
| <b>Vel</b>    | Current speed, in inches/minute (0 when stopped)                          |
| <b>End</b>    | Why the last move ended: zero, hardzero, stepdone, or limit               |
| <b>LimMax</b> | Longest unbroken "pressed" reading seen on the limit line this run, in ms |

The **End** and **LimMax** fields are troubleshooting aids. A limit trip fires only after the switch reads pressed for an unbroken ~40 ms. If LimMax keeps creeping toward 40 ms during a normal run (with nothing touching the switch), electrical noise is getting close to the trip threshold — see Troubleshooting.

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## 12. Safety features and automatic behaviors

- **Homing on startup:** every power-on finds home via the limit switch, so the position reference is always correct.
- **Hard zero clamp:** the ghost physically cannot travel past home (position zero). Reverse jog is blocked there.
- **Limit switch E-stop:** if the limit switch is pressed during a run, the motor stops immediately (handled in the motor's interrupt routine so it reacts even at high speed) and the run turns **Off**. A banner shows how long the switch was held.
- **Triple-trigger "go home":** tapping/triggering the limit switch **3 times within 3 seconds** — whether running or idle — sends the ghost back to home and parks it. Handy as a manual "reset/park" gesture.
- **Noise hardening:** the limit switch and the button both use multi-sample confirmation so motor electrical noise doesn't cause false triggers.
- **Screensaver:** after **1 minute** of no input, a bouncing-ghost animation with the clock appears. While it's showing, the **button is ignored** (so stray noise during a long unattended run can't change settings). **Turn the knob** to wake it; the first turn only wakes the screen (it won't also move the menu).
- **Power-loss memory:** positions, speeds, mode, run state, schedule, and clock are all stored and restored automatically.

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## 13. Typical first-run checklist

1. Flash the firmware (Section 3).
2. Power on. Let it **home** automatically.

3. It enters **Learn mode** — jog to **A**, click; jog to **B**, click.
4. Go to **Setup > Set Clock** and set the correct time.
5. (Optional) **Setup > Set Schedule** to set a nightly window.
6. Go to **Mode** and pick **Fwd-Back** or **Wander**.
7. Open **Motion** and set Speed, Accel/Decel, and Dwell to taste.
8. Highlight **Run** and click to choose **Continuous** or **Schedule**.
9. Watch it go. Use **Diagnostics** to confirm position and speed look right.

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## 14. Troubleshooting

| Symptom                                                       | Likely cause / fix                                                                                                                   |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Distances/speeds look wrong (e.g. "feet" don't match reality) | DIP switch microstepping isn't 1/8. Set MS1/MS2/MS3 = H/H/L, or change STEPS_PER_INCH in firmware.                                   |
| "Set positions first!" when selecting Run                     | You haven't taught A and B yet. Do Setup > Learn Positions.                                                                          |
| "RTC not found!" in Schedule/Clock                            | DS3231 not detected on I2C. Check wiring/address (0x68). Schedule needs the RTC.                                                     |
| Display never comes on                                        | OLED not detected at I2C 0x3C. The onboard LED will blink rapidly as an error signal. Check SDA/SCL wiring.                          |
| Ghost stops mid-run unexpectedly and run turns Off            | Limit switch tripped (real press or noise). Check Diagnostics End=limit and LimMax. If noise, improve limit-switch wiring/shielding. |
| Reverse jog won't move in Learn mode                          | You're at home (position zero); reverse past home is blocked by design.                                                              |
| Button feels unresponsive during a run                        | Expected: the step pulses briefly starve the loop at high speeds. A deliberate tap+release (held $\geq$ ~40 ms) still registers.     |
| Screensaver won't respond to clicks                           | By design — the button is deaf during the screensaver. <b>Turn the knob</b> to wake it.                                              |
| Mode didn't switch right away                                 | If mid-move, the switch is deferred until the current leg finishes ("Mode switch after move").                                       |

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## 15. Quick reference card

TURN knob = move highlight / change value  
 CLICK knob = select / enter / confirm

Main menu: Run | Mode | Motion | Setup | Diagnostics

Run cycle: Off -> Schedule -> Continuous -> Off

Modes: Fwd-Back (fixed dwell) | Wander (random target + random dwell)

Learn: turn CW = forward, CCW = reverse, click = save (A then B)  
Edit a setting: click to edit (\*), turn to change, click to save  
Park now: trigger limit switch 3x within 3 seconds -> goes home  
Wake screensaver: turn the knob